Cross-Currency Basis Swaps

Pricing, Quoting Conventions, and Cross-Gamma in the MTM

A desk primer covering float–float, fixed–float, fixed–fixed and float–fixed structures, the basis and its conventions on non-USD pairs, and the FX–rates convexity in mark-to-market

Working reference · multi-curve / OIS-discounting conventions assumed throughout

1 What a cross-currency swap is

A cross-currency (XCCY) swap exchanges cashflows in *two different currencies*. Unlike a single-currency IRS, the two legs are in different units of account, so

- **notionals are exchanged** at inception and re-exchanged at maturity (this principal exchange is the defining feature — it is what creates the funding/FX linkage), and
- the two notionals are set at the *inception spot* FX rate, so $N_{\text{for}} = N_{\text{dom}}/S_0$ (quoting S_0 as domestic units per one foreign unit, or the appropriate market convention for the pair).

Because you can fund in one currency and swap into another, the XCCY swap is the instrument that enforces *covered interest parity* — and the persistent deviation from textbook CIP is exactly the **cross-currency basis**.

1.1 The four leg combinations

Structure	Leg A	Leg B
Float–float (basis swap)	foreign floating + basis	USD floating (flat)
Fixed–float	foreign fixed	USD floating
Float–fixed	foreign floating + basis	USD fixed
Fixed–fixed	foreign fixed	USD fixed

The market *quotes* the float–float form; the other three are obtained by swapping one or both floating legs for their fixed equivalents using the relevant single-currency swap rate. All four price off the same two-currency, basis-adjusted discounting framework below.

2 The multi-curve, dual-currency pricing framework

2.1 Curves you need

For a pair (foreign "f", USD "$"), pricing requires:

1. a USD discount curve $P_{\$}(0,t)$ (USD OIS, e.g. SOFR),
2. a USD projection curve for the USD floating index,
3. a foreign projection curve for the foreign floating index, and
4. a *basis-adjusted* foreign discount curve $P_f(0,t)$, built so that a par XCCY basis swap prices to zero.

Key idea. You do *not* discount foreign cashflows on the plain foreign OIS curve. You discount them on a curve that has been shifted by the cross-currency basis, so that the market-quoted basis swap is worth zero at inception. The basis lives *in the discounting* of the non-USD leg.

2.2 Par condition for the float–float basis swap

Let the foreign leg pay its floating index plus a constant basis spread b, and the USD leg pay USD floating flat. Converting both legs to a common currency (USD) at inception spot S_0, the par condition "PV $= 0$" that *defines* the basis-adjusted foreign discount curve is, schematically,

$$\underbrace{S_0\Big[\,\mathrm{PV}^{f}_{\mathrm{flt}} + b\sum_i \tau_i P_f(0,T_i)\,N_f + \text{principal exch.}\Big]}_{\text{foreign leg in USD}} \;=\; \underbrace{\mathrm{PV}^{\$}_{\mathrm{flt}} + \text{principal exch.}}_{\text{USD leg}}.$$

Including principal exchange, the foreign leg PV (in foreign units, on the basis-adjusted curve) telescopes to

$$\mathrm{PV}^{f}_{\mathrm{leg}} = \underbrace{\big(P_f(0,T_0) - P_f(0,T_m)\big)N_f}_{\text{floating + principals}} + b\,N_f \sum_{i=1}^{m} \tau_i P_f(0,T_i).$$

Setting the USD-converted total to zero and solving for the curve (or, given the curve, for b) is the bootstrap. The basis b is the number quoted in the market.

2.3 Why the basis is nonzero: CIP deviation

Under textbook covered interest parity the forward FX rate satisfies

$$F_{0,T} = S_0\,\frac{P_f(0,T)}{P_{\$}(0,T)} \quad \text{(no basis)}.$$

In practice supply/demand for dollar funding, balance-sheet costs and regulatory frictions drive a wedge, so the traded forward implies a foreign discount curve shifted by b. A negative EUR/USD basis, for example, means it is *expensive* to borrow dollars via the FX market — non-US entities pay up (receive a negative spread on the EUR leg) to obtain USD.

3 Quoting conventions (non-USD pairs)

3.1 USD is (almost always) the flat leg

By convention the **USD leg is quoted flat** and the **basis spread is attached to the non-USD leg**. So "EUR/USD 5Y basis at -12" means: pay 3M USD LIBOR/SOFR flat versus receive 3M EURIBOR/STR *minus* 12 bp (or vice versa). The sign is expressed on the non-USD leg.

3.2 Index and frequency conventions by currency

Pair	Legacy convention	RFR-era convention
EUR/USD	3M EURIBOR vs 3M USD LIBOR	STR vs SOFR (basis on STR leg)
GBP/USD	3M GBP LIBOR vs 3M USD LIBOR	SONIA vs SOFR
JPY/USD	3M JPY LIBOR (TIBOR) vs USD	TONA vs SOFR

Practical points that trip people up:

- **Quotation direction.** Market pairs are quoted a specific way (EUR/USD as USD per EUR, USD/JPY as JPY per USD). Get S_0 the right way up or every sign flips.
- **Notional resets (MTM vs non-MTM).** Most interbank XCCY basis swaps are *mark-to-market* ("resetting notional"): the USD notional is reset to prevailing spot at each period, with a compensating principal payment. This strips out a large part of the FX exposure between resets. *Non-MTM* (constant-notional) swaps leave the full FX exposure on — and, as we will see, a different convexity profile.

- **Collateral / CSA currency.** Discounting must match the CSA currency. A USD-CSA trade discounts *all* legs at USD OIS (via the FX-forward-implied curve for the foreign leg); a mismatched CSA changes the basis you back out.
- **RFR transition.** Post-LIBOR, legs reference compounded RFRs; the basis is now quoted RFR-vs-RFR (e.g. STR/SOFR) and a separate LIBOR-fallback / index basis may appear on legacy trades.

4 Pricing the four structures

All four use the same discount/projection curves; they differ only in what each leg pays. Write foreign-leg and USD-leg PVs (each in its own currency) and convert the foreign leg at spot S_0 for a common-currency PV.

4.1 Float–float (the quoted basis swap)

$$\mathrm{PV} = S_0\Big[\big(P_f(0,T_0) - P_f(0,T_m)\big)N_f + bN_f\sum_i \tau_i^f P_f(0,T_i)\Big] - \Big[\big(P_\$(0,T_0) - P_\$(0,T_m)\big)N_\$\Big].$$

At par with market b, PV $= 0$. This is the calibration instrument.

4.2 Fixed–float

Replace the foreign floating+basis leg with a foreign *fixed* coupon K_f:

$$\mathrm{PV} = S_0\Big[K_f N_f\sum_i \tau_i^f P_f(0,T_i) + \big(P_f(0,T_0) - P_f(0,T_m)\big)N_f\Big]_{\text{principals}} - \big[\text{USD floating leg}\big].$$

The fair K_f is the foreign fixed rate that makes this zero *on the basis-adjusted foreign curve* — i.e. it already embeds the basis, and therefore differs from the plain domestic swap rate.

4.3 Float–fixed

Foreign floating+basis versus *USD fixed* $K_\$$:

$$\mathrm{PV} = S_0\big[\text{foreign float} + b\big] - \Big[K_\$ N_\$\sum_i \tau_i^\$ P_\$(0,T_i) + \text{USD principals}\Big].$$

4.4 Fixed–fixed

Both legs fixed — common for issuers swapping a fixed-rate bond in one currency into fixed in another:

$$\mathrm{PV} = S_0\Big[K_f N_f\sum_i \tau_i^f P_f(0,T_i) + \text{princ.}\Big] - \Big[K_\$ N_\$\sum_i \tau_i^\$ P_\$(0,T_i) + \text{princ.}\Big].$$

> **The basis is currency-of-discounting, not a coupon add-on.** Whichever structure you trade, the cross-currency basis enters through the *basis-adjusted foreign discount curve*. Converting a quoted float–float basis into an equivalent fixed–fixed all-in rate is just re-expressing the same discounting on fixed coupons.

5 Non-USD crosses (EUR/GBP, EUR/JPY, . . .): deriving the basis

Everything above assumed one leg is USD. But a large share of real trades are *non-USD vs non-USD* — EUR/GBP, EUR/JPY, EUR/CHF, and so on. These have **no USD leg at all**, so "USD flat with the spread on the other leg" does not apply. The question your team is right to ask is: where does the basis come from, and how do the MTM resets work when neither notional is USD?

5.1 USD is the pivot currency

The key structural fact of the XCCY market is that **liquidity is concentrated in $-crosses**. EUR/USD, GBP/USD, JPY/USD basis are deeply quoted; EUR/GBP basis is comparatively thin and, where it trades, is largely kept consistent with the two dollar legs by arbitrage. So the non-USD basis is almost always *constructed by triangulation through USD* rather than taken from a primary quote.

5.2 Triangulation: the cross basis is a difference of two USD bases

Think of building a synthetic EUR/GBP basis swap out of two USD basis swaps:

1. Pay EUR floating, receive USD floating $+\, b_{\mathrm{EUR/USD}}$ (a EUR/USD basis swap), and
2. Pay USD floating, receive GBP floating $+\, b_{\mathrm{GBP/USD}}$ (a GBP/USD basis swap).

The two USD legs are equal and opposite, so they **net to zero**, leaving a synthetic EUR-vs-GBP basis swap. To first order the EUR/GBP basis is therefore the *difference* of the two dollar bases:

$$b_{\mathrm{EUR/GBP}} \;\approx\; b_{\mathrm{GBP/USD}} - b_{\mathrm{EUR/USD}},$$

expressed on the appropriate leg. Intuitively, each currency's "expensiveness of dollars" is measured against USD; the cross basis is what is *left over* once the common USD funding component cancels.

> **The non-USD basis is a residual.** EUR/GBP basis ≈ (GBP/USD basis) − (EUR/USD basis). It is not independently quoted so much as *implied* by the two dollar legs and kept in line by arbitrage. This is why a move in USD funding stress shows up in *every* cross simultaneously.

5.3 Curve construction: discount off the pivot, not off spot parity

Operationally you do *not* build a standalone EUR/GBP basis-adjusted curve from a direct quote. Instead:

1. Build each currency's USD-basis-adjusted discount curve from its $-cross (EUR discounted consistently with EUR/USD basis; GBP with GBP/USD basis), all anchored to the same USD OIS / CSA base.
2. Price the EUR/GBP swap by converting each leg into the common base (USD, or whichever the CSA specifies), using those already-basis-adjusted curves.

Because both foreign curves are pinned to the same USD base, the EUR/GBP basis *falls out* of the two curves automatically — there is no separate calibration step, and no need for a direct EUR/GBP quote. The cross FX rate is itself the ratio $S_{\mathrm{EUR/GBP}} = S_{\mathrm{EUR/USD}}/S_{\mathrm{GBP/USD}}$, consistent with the same triangulation.

5.4 Which leg carries the spread, and sign conventions

With no USD leg there is no automatic "flat" side, so the pair has its own market convention for which leg the spread sits on (often the currency that is the "base" of the market quote, e.g. the spread on the EUR leg for EUR/GBP). The safe habit: always confirm (i) which leg the spread is attached to, (ii) the sign, and (iii) the FX quote direction, because triangulated signs flip easily. A booking error here typically shows up as a basis DV01 with the wrong sign against both dollar legs.

5.5 MTM / resetting notional when neither leg is USD

The mark-to-market (resetting-notional) mechanic still applies, but with a twist:

- **You must choose the reset (base) currency.** In a $-cross the USD notional resets to spot. In EUR/GBP one leg is nominated as the constant ("base") notional and the *other* resets to the prevailing EUR/GBP spot, with the compensating principal paid in the reset currency. Which side resets is a convention/negotiation, not a given — and it determines which currency's discount factors the reset flows land in.
- **The reset FX is the cross, so its vol is a triangulated vol.** The EUR/GBP spot used at each reset has volatility $\sigma^2_{\text{EUR/GBP}} = \sigma^2_{\text{EUR/USD}} + \sigma^2_{\text{GBP/USD}} - 2\rho\,\sigma_{\text{EUR/USD}}\sigma_{\text{GBP/USD}}$. The reset convexity (Effect 2 in the next section) therefore depends on the *correlation between the two dollar rates*, not on a single clean FX vol.
- **Residual USD sensitivity.** Even though no cashflow is in USD, the *valuation* runs through USD-anchored curves, so the MTM retains a (usually small) sensitivity to each currency's USD basis. A shock to EUR/USD basis alone moves a EUR/GBP book. Desks that think of EUR/GBP as "dollar-free" are surprised by this; it is the triangulation showing through.

Take-away for the cross. A EUR/GBP basis swap is best understood as *two dollar basis swaps with the USD legs cancelled*. Its basis, its curve, its reset FX and its convexity are all triangulated quantities — so its risk lives not just in the EUR/GBP basis pillar but also, residually, in both the EUR/USD and GBP/USD basis ladders and in the EUR/USD–GBP/USD correlation.

6 Mark-to-market and where the convexity comes from

This is the part your team is discussing: *when you MTM the non-USD leg, there is convexity.* There are two distinct effects, and it is worth separating them.

6.1 Effect 1 — FX–rates cross-gamma (the main one)

Your book reports PV in a base/reporting currency (say USD). The foreign leg's PV is naturally computed in foreign units, V_f, then converted at spot S:

$$\Pi(S, \mathbf{r}_f) \;=\; S \cdot V_f(\mathbf{r}_f),$$

where $\mathbf{r}_f$ denotes the foreign rates/curve. **PV is a product of two risk factors**: the FX rate S and the rates-driven PV V_f. A product has a *cross second derivative* even though it has no "own" curvature in either variable alone:

$$\frac{\partial^2 \Pi}{\partial S\,\partial \mathbf{r}_f} \;=\; \frac{\partial V_f}{\partial \mathbf{r}_f} \;\neq\; 0.$$

The second-order Taylor expansion of the MTM makes the cross term explicit:

$$\Delta\Pi \;\approx\; \underbrace{V_f\,\Delta S}_{\text{FX delta}} + \underbrace{S\,\frac{\partial V_f}{\partial \mathbf{r}_f}\Delta \mathbf{r}_f}_{\text{rate delta}} + \underbrace{\frac{\partial V_f}{\partial \mathbf{r}_f}\,\Delta S\,\Delta \mathbf{r}_f}_{\textbf{cross-gamma}} + \tfrac{1}{2} S\,\frac{\partial^2 V_f}{\partial \mathbf{r}_f^2}(\Delta \mathbf{r}_f)^2 + \cdots$$

This is the convexity in the non-USD leg's MTM. The cross-gamma term $\frac{\partial V_f}{\partial \mathbf{r}_f}\,\Delta S\,\Delta \mathbf{r}_f$ says: your *effective rate DV01 in reporting currency scales with the FX rate.* When spot moves, your foreign-rate delta (measured in base currency) changes — so a hedge put on at one spot level is mis-sized once spot moves. Equivalently, your FX delta V_f changes as foreign rates move. The two sensitivities are coupled; that coupling is the cross-gamma.

Sign and hedging intuition. If foreign rates rally and V_f rises *while* the foreign currency strengthens, the base-currency PV gains on both counts and the cross term *adds* — a favourable co-movement is convex in your favour; an adverse anti-correlated move is concave against you. Because the cross-gamma depends on the realised *correlation* between FX and foreign rates, it cannot be hedged with static FX or static rate delta alone — you must either re-hedge dynamically (delta-rebalance as spot moves) or hold offsetting FX/rates optionality. This is why desks watch the FX–rates correlation in their XCCY books.

6.2 Effect 2 — resetting-notional (MTM) swap convexity

In a *mark-to-market* XCCY swap the USD notional is reset to prevailing spot at each reset, with a compensating notional payment $\left(N_{\$}^{\text{new}} - N_{\$}^{\text{old}}\right)$ exchanged. That compensating flow is itself *spot-dependent*, so each reset introduces a small payoff that is a function of S at the reset date:

$$\text{reset flow}_k \;\propto\; \left(S_{t_k} - S_{t_{k-1}}\right) N_f.$$

Marking these to market embeds a strip of forward-FX-linked flows whose value depends on the *covariance of FX with the discount factors* — a convexity ("quanto-like") adjustment distinct from Effect 1. Non-MTM swaps do not have these reset flows, but carry larger outright FX exposure between the initial and final principal exchange instead. *For a non-USD cross* the relevant ΔS is the cross-FX move, whose variance is the triangulated $\sigma^2_{\text{EUR/GBP}} = \sigma^2_{\text{EUR/USD}} + \sigma^2_{\text{GBP/USD}} - 2\rho\,\sigma_{\text{EUR/USD}}\sigma_{\text{GBP/USD}}$, so the reset convexity is driven by the correlation between the two dollar pairs (§6).

6.3 Effect 3 — quanto / correlation adjustment on the floating fixings

More subtly, the *expected* foreign floating rate used to value the leg is taken under the USD (or CSA-currency) risk-neutral measure once you discount in USD. Changing measure from the foreign to the domestic measure introduces a drift/convexity adjustment proportional to the covariance between the foreign forward rate and the FX rate:

$$\mathbb{E}^{\$}\big[F_i^f\big] \;\approx\; \mathbb{E}^{f}\big[F_i^f\big] \;+\; \underbrace{\rho\,\sigma_F\,\sigma_S\,(\cdots)}_{\text{quanto/convexity adj.}},$$

with ρ the FX–rate correlation. This is second-order for most vanilla flow but becomes material for long-dated or high-vol pairs, and it is the rigorous origin of "there is convexity when you MTM the foreign leg."

7 Risk representation and hedging

- **Three delta axes.** An XCCY book carries (i) foreign-rate DV01, (ii) USD-rate DV01, and (iii) FX delta — plus (iv) **basis DV01** (sensitivity to the cross-currency basis curve itself), which is its own pillar ladder.
- **Cross-gamma is a matrix, not a number.** The relevant second-order risk is the mixed partial FX×foreign-rate (and FX×basis). Report it as a cross-gamma grid: how does foreign-rate DV01 (in base ccy) shift per 1% move in spot.

- **Dynamic rehedging.** Because the base-currency rate delta scales with spot, rate hedges must be re-sized as spot moves; conversely FX hedges must be re-sized as foreign rates move. The rebalancing cost is the P&L cost of the cross-gamma.
- **Basis ladder.** Just like the single-curve DV01 ladder, XCCY basis risk is bucketed to the quoted basis pillars (1Y, 2Y, ..., 30Y). Hedge with market basis swaps at those tenors; reconcile at pillar level.
- **CSA consistency.** Ensure the discounting currency matches the CSA; otherwise the basis you extract — and hence every downstream sensitivity — is mis-specified.

8 Summary

- An XCCY swap exchanges cashflows in two currencies *with principal exchange*; the market quotes the float–float form with USD flat and the basis on the non-USD leg.
- The cross-currency basis enters through a **basis-adjusted foreign discount curve** calibrated so the par basis swap prices to zero — it is a deviation from covered interest parity.
- Fixed–float, float–fixed and fixed–fixed are the same framework with one or both floating legs replaced by their fixed equivalents; the fair fixed rate embeds the basis.
- Conventions to nail: quote direction of spot, USD-flat quoting, MTM vs non-MTM notional resets, CSA/discounting currency, and RFR-era index basis.
- **Convexity in the MTM of the non-USD leg** has three sources: (1) *FX–rates cross-gamma* because base-currency PV is $S \times V_f$, a product of two risk factors; (2) *resetting-notional* reset flows in MTM swaps; (3) the *quanto/measure-change* adjustment on foreign fixings. All three are governed by the FX–foreign-rate correlation and require dynamic rehedging rather than static delta.

Notation: $P_c(0,t)$ discount factor in currency c; S spot FX (domestic per foreign, per market convention); b cross-currency basis spread; V_f foreign-leg PV in foreign units; N_c notional in currency c; ρ FX–rate correlation. Multi-curve, CSA-consistent (OIS) discounting assumed.